

Slime Hand XR: Distinct Illusory Skin Deformation in HMD Space

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Figure 1: SLIME HAND XR

Abstract

Slime Hand XR is an interactive head-mounted display (HMD) system that induces illusory skin deformation in VR. In the proposed system, participants wearing the HMD place their hands on a tray where a chunk of slime is visible within the HMD's view. Participants can instantly and boldly feel illusory skin stretching by looking at the deformed slime while the experimenter pinches and pulls their hand's skin at the same time. In addition, the tip of the deformed slime can be clipped with a clothespin in VR by the experimenter. Then, this setup can maintain the form of illusory skin deformation. This system adopts the principle of the slime hand illusion, which can induce a strong sense of skin deformation. By using computer-generated slime, we achieved a unique skin deformation illusion that is hard to replicate in reality, such as suspended deformation. In testing the illusion effect of this system at

an open laboratory exhibition for children, 56 out of 63 participants reported a strong reaction to the sensation of skin deformation in the questionnaire.

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1 Background

The report of body ownership illusion, such as the rubber hand illusion [Botvinick and Cohen 1998] (RHI) shows that the body image has high plasticity. There are various proposals related to body image deformation using the RHI paradigm in VR [Kiltner et al. 2012][Kodaka and Mori 2017]. However, as is well known in the psychological studies, RHI tend to be less likely to occur when the fake hand the real hand are more than 20cm apart. This is because RHI is constrained by limitations derived from proprioception. In considering the future development of products, it is preferable to adopt an illusion principle that avoids the limitations of RHI while allowing flexible deformation when using body-deforming avatars.

The Slime Hand Illusion (SHI) reported by Kodaka et al. in 2022 is an illusion that induces an excessive sensation of skin deformation.

SHI demonstrated a drift in the sense of skin location, which indicates that SHI avoid the proprioceptive constraints of traditional RHI[Kodaka et al. 2022]. Such remarkable malleability of perceived skin is believed to occur because the skin cannot perceive positional information in the same way as proprioceptive receptors.

In this paper, we replicate the illusory effect of SHI in a VR space using an HMD and propose a body deformation sensation system called 'Slime Hand XR', which circumvents the limitations by proprioception. In Slime Hand XR, the SHI effect is recreated by placing computer-generated slime that stretches and contracts at the position of the participant's pinched skin. By using computer-generated graphics (CG) to simulate the skin deformation effect of SHI, we can achieve flexible forms of skin deformation that cannot be achieved using analog slime (e.g. suspended skin deformation).

2 Interaction Design

The HTC Vive was adopted as the HMD to induce the sensation of skin deformation. The participant view a room through the HMD's viewpoint, where a tray (with a chunk of slime in the middle) and a hanger with a clothespin are visible. In this system, three HTC Vive trackers are used to represent the deformation of the CG slime. Tracker (a) tracks the position of the tip of the CG slime; Tracker (b), which is placed on the tray in reality, tracks the position of the participant's hand; Tracker (c) tracks the position of the hanger with a clothespin.

2.1 Lifted deformation mode

In Slime Hand XR, the participant wearing the HMD places their hand on the center of the tray, where their skin is pinched and pulled by the experimenter. During this process, the experimenter moves Tracker (a) to manipulate the stretching of the CG slime. Then, the participant feel a sensation of skin deformation.

2.2 Suspended deformation mode

When the experimenter brings Tracker (a) close to Tracker (c), which indicates the hanger's position in VR, the clothespin on the hanger automatically pinches the CG slime. After this, The role of Tracker (a) is handed over to Tracker (c). Then, the tip of the CG slime is suspended in mid-air, pinched by the clothespin. With the tip of CG slime suspended by the clothespin, the participant can manipulate its stretching and contracting by moving the tray with their free hand. Simultaneously, the experimenter can adjust the position of the CG slime's tip, pinched by the clothespin, by manipulating Tracker (c).

3 EVALUATIONS

We conducted subjective evaluations using a questionnaire regarding the skin deformation sensation in Slime Hand XR. Participants evaluated the skin deformation effect on a 7-point scale (-3 to +3). When this system was in the prototype stage, in testing the illusion effect of this system at an open laboratory exhibition for children, 56 out of 63 participants (89%) reported a strong reaction (rated +2 or higher) in the questionnaire. The current system was briefly tested in our laboratory experiment. The result showed that 19 out of 20 participants (95%) agreed positively with the effect of Lifted Deformation mode (rated +1 or more), and 13 participants (65%)

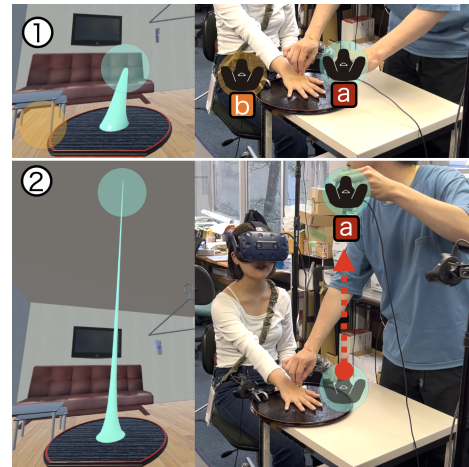


Figure 2: Lifted deformation mode

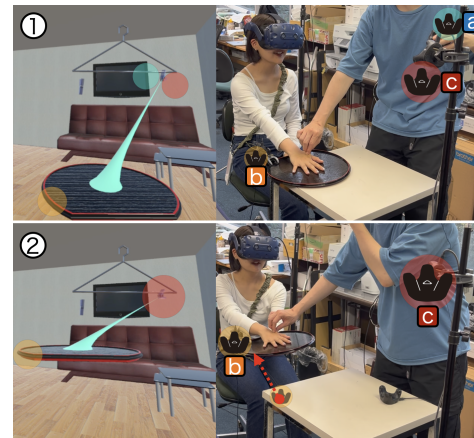


Figure 3: Suspended deformation mode

agreed positively with the effect of Suspended deformation mode. The result suggest that our system successfully induced the unique sensation of skin deformation achievable only in VR.

References

- M Botvinick and J Cohen. 1998. Rubber hands 'feel' touch that eyes see. *Nature* 391, 6669 (feb 1998), 756. <https://doi.org/10.1038/35784>
- Konstantina Kiltani, Jean Marie Normand, Maria V. Sanchez-Vives, and Mel Slater. 2012. Extending body space in immersive virtual reality: A very long arm illusion. *PLoS ONE* 7 (2012), Issue 7. <https://doi.org/10.1371/journal.pone.0040867>
- K. Kodaka and K. Mori. 2017. Stretchchar(m) makes your arms elastic. *SIGGRAPH Asia 2017 VR Showcase, SA 2017*. <https://doi.org/10.1145/3139468.3139476>
- Kenri Kodaka, Yutaro Sato, and Kento Imai. 2022. The slime hand illusion: Non-proprioceptive ownership distortion specific to the skin region. 13 (11 2022), 204166952211377. Issue 6. <https://doi.org/10.1177/20416695221137731>